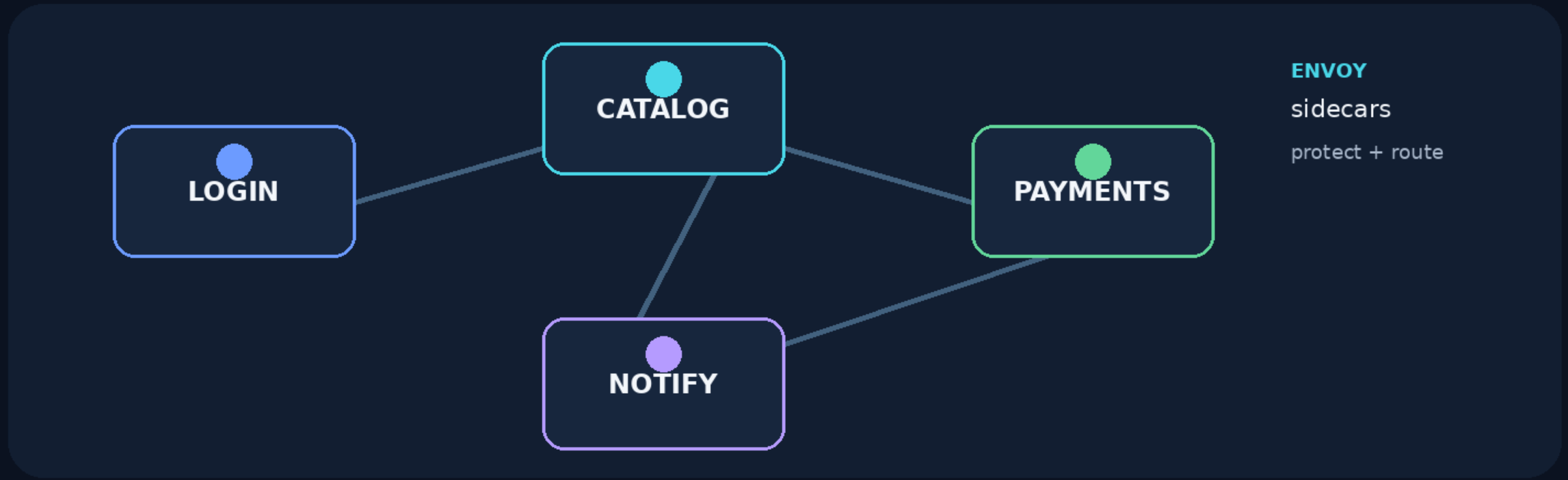
KUBERNETES NETWORKING

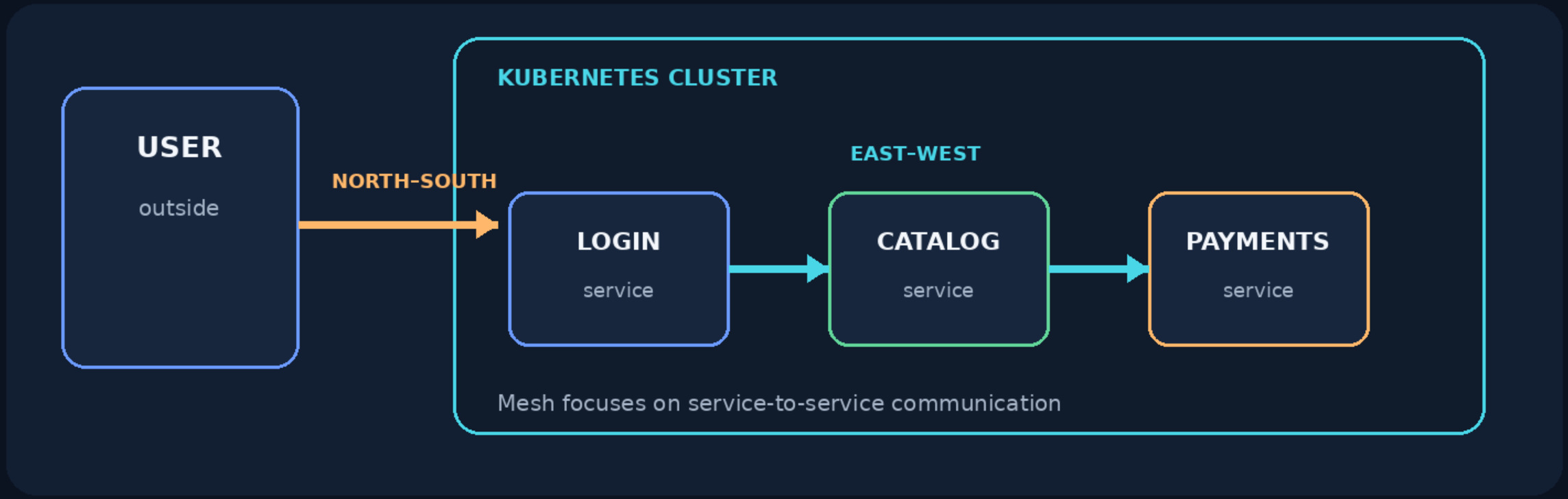
Istio & Service Mesh

mTLS • Sidecar injection • Canary releases • Observability



A mesh manages east-west traffic

North-south enters/leaves the cluster. East-west moves between workloads.



Move cross-cutting logic out of app code

Consistent security, traffic policy and telemetry across services.

IDENTITY + mTLS

Encrypt and authenticate workloads

TRAFFIC CONTROL

Canary, retries, timeouts, circuit breaking

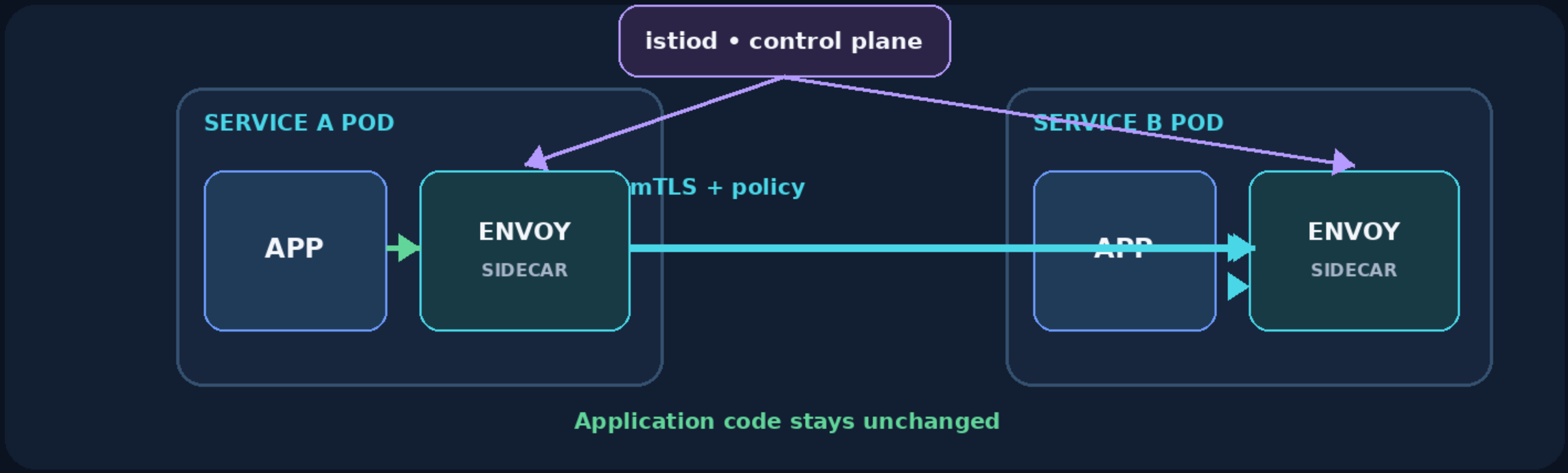
OBSERVABILITY

Metrics, service graph and traces

Trade-off: proxy resources, added latency and operational complexity.

The sidecar sits beside each app

Envoy intercepts workload traffic; istiod distributes configuration and certificates.



How does the sidecar get there?

Automatic injection is performed at Pod admission by Istio's mutating webhook.



Namespace label: istio-injection=enabled • New Pods are injected; existing Pods need recreation.

mTLS: both ends prove identity

PeerAuthentication controls what inbound traffic a workload accepts.

PERMISSIVE

mTLS or plaintext

STRICT

mTLS only

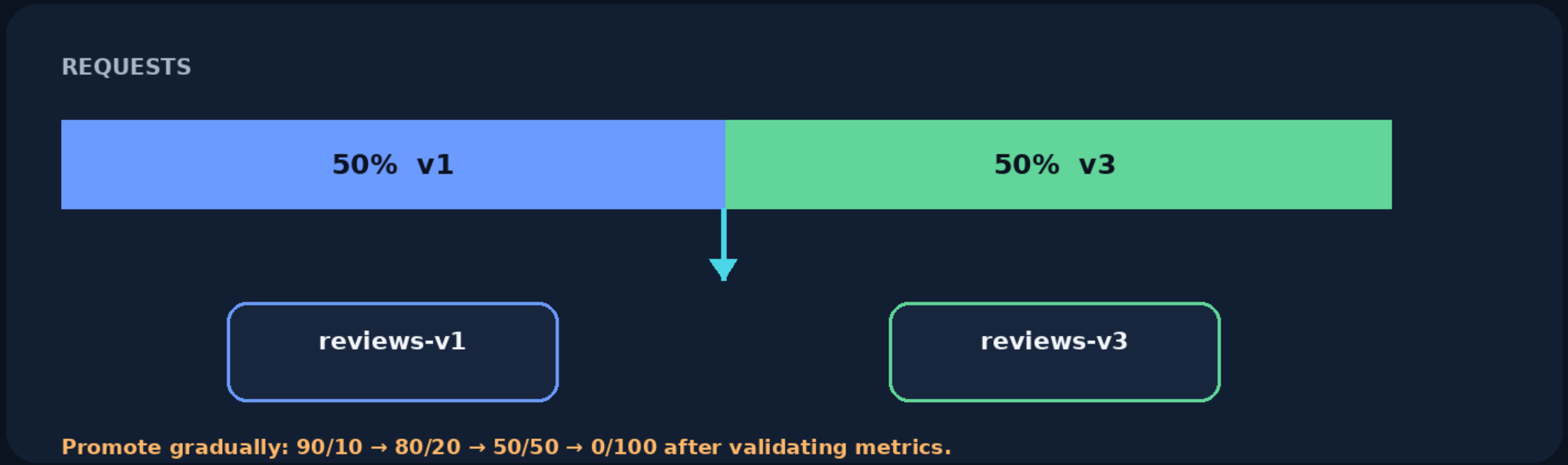
DISABLE

No Istio mTLS

Client Envoy ⇄ certificates + encrypted channel ⇄ Server Envoy

Canary releases without app changes

DestinationRule defines subsets. VirtualService assigns traffic weights.



Promote gradually: 90/10 → 80/20 → 50/50 → 0/100 after validating metrics.

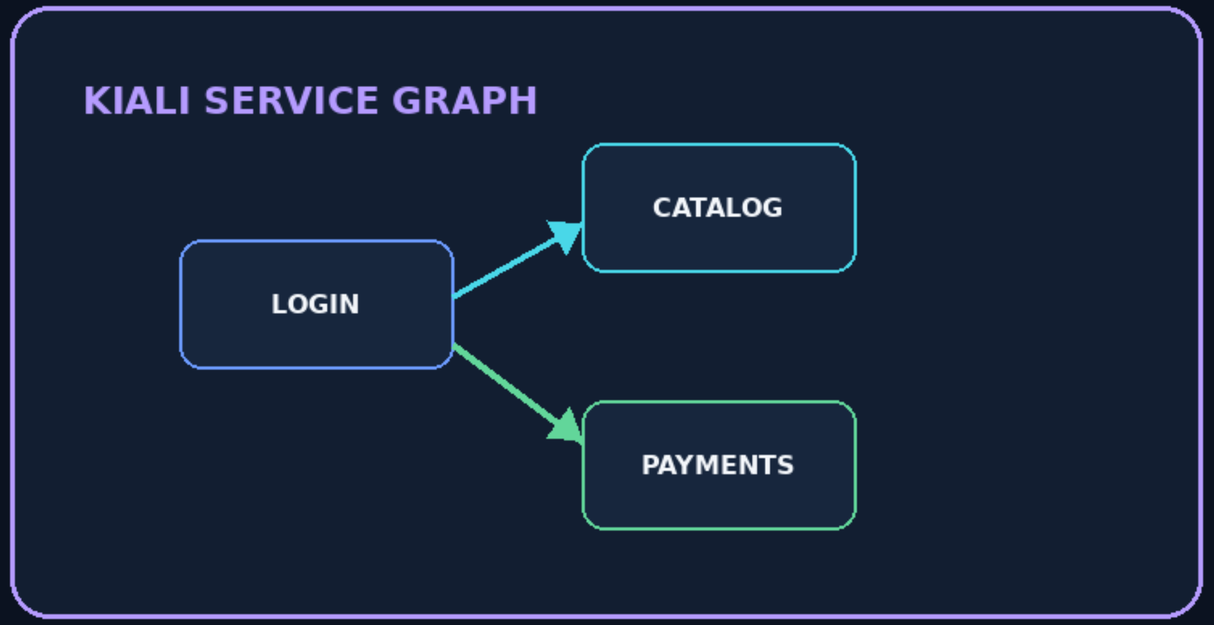
Bookinfo: see a mesh in action

Four microservices; reviews has three versions, making traffic changes visible.



Observe, diagnose, improve

Kiali visualises service relationships using telemetry from the mesh.



QUICK CHECKS

- ✓ Pods 2/2? Injection label present?
- ✓ Subset labels match DestinationRule?
- ✓ VirtualService host + weights correct?
- ✓ Kiali add-ons installed and traffic generated?

Understand the moving parts

A practical mental model for interviews and real clusters.

01 Envoy proxies carry data-plane traffic

02 istiod configures proxies and manages identity

03 PeerAuthentication sets inbound mTLS policy

04 DestinationRule + VirtualService shape traffic

05 Kiali helps inspect service relationships

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